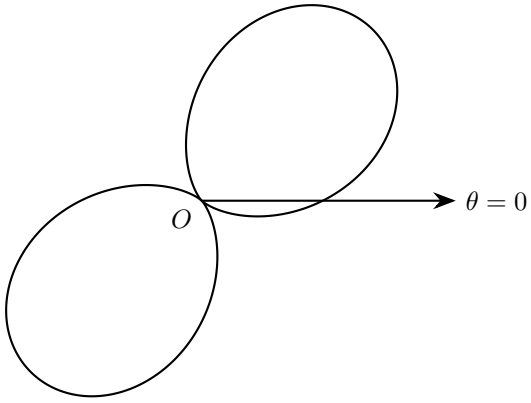
CORE PURE / POLAR COORDINATES

Integration of Polar Curves

Worked solutions 12 questions · 96 marks

Questions

Question 1	*9 marks*	2
Question 2	*10 marks*	5
Question 3	*7 marks*	7
Question 4	*6 marks*	9
Question 5	*8 marks*	12
Question 6	*8 marks*	15
Question 7	*8 marks*	17
Question 8	*7 marks*	20
Question 9	*8 marks*	23
Question 10	*8 marks*	26
Question 11	*7 marks*	28
Question 12	*10 marks*	31



1. The diagram above shows the curve with polar equation $r = 1 + \sin 2\theta$ for $0 \leq \theta \leq 2\pi$.

(a) Find the values of θ at the pole. **[1]**

(b) Find the polar coordinates of the points on the curve where r takes its maximum value. **[2]**

(c) Find the exact total area enclosed by the curve. **[4]**

(d) Find a Cartesian equation for the curve. **[2]**

Solution

(a) At the pole, $r = 0$. So

$$1 + \sin 2\theta = 0$$

$$\sin 2\theta = -1$$

For $0 \leq \theta \leq 2\pi$, this gives

$$2\theta = \frac{3\pi}{2}, \ \frac{7\pi}{2}$$

so

$$\theta = \frac{3\pi}{4}, \ \frac{7\pi}{4}$$

Therefore the values of θ at the pole are $\frac{3\pi}{4}$ and $\frac{7\pi}{4}$.

(b) $r = 1 + \sin 2\theta$ is largest when $\sin 2\theta = 1$. Hence

$$r_{\max} = 1 + 1 = 2$$

Now $\sin 2\theta = 1$ when

$$2\theta = \frac{\pi}{2}, \ \frac{5\pi}{2}$$

so

$$\theta = \frac{\pi}{4}, \ \frac{5\pi}{4}$$

Therefore the points are

$$(2, \tfrac{\pi}{4}) \quad \text{and} \quad (2, \tfrac{5\pi}{4})$$

(c) The total area enclosed by a polar curve is

$$A = \frac{1}{2}\int_0^{2\pi} r^2 \, \mathrm{d}\theta$$

So here

$$A = \frac{1}{2}\int_0^{2\pi} (1 + \sin 2\theta)^2 \, \mathrm{d}\theta$$

Expanding,

$$A = \frac{1}{2}\int_0^{2\pi} \left(1 + 2\sin 2\theta + \sin^2 2\theta\right) \mathrm{d}\theta$$

Using

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

gives

$$\begin{aligned} A &= \frac{1}{2}\int_0^{2\pi} \left(1 + 2\sin 2\theta + \frac{1 - \cos 4\theta}{2}\right) \mathrm{d}\theta \\ &= \frac{1}{2}\int_0^{2\pi} \left(\frac{3}{2} + 2\sin 2\theta - \frac{1}{2}\cos 4\theta\right) \mathrm{d}\theta \end{aligned}$$

Integrating term by term,

$$\begin{aligned} A &= \frac{1}{2}\left[\frac{3}{2}\theta - \cos 2\theta - \frac{1}{8}\sin 4\theta\right]_0^{2\pi} \\ &= \frac{1}{2}(3\pi - 1 - 0 - (-1)) \\ &= \frac{3\pi}{2} \end{aligned}$$

(d) Using

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r^2 = x^2 + y^2$$

and $\sin 2\theta = 2\sin\theta\cos\theta$,

$$r = 1 + 2\sin\theta\cos\theta$$

Now

$$\sin\theta = \frac{y}{r}, \qquad \cos\theta = \frac{x}{r}$$

so

$$r = 1 + \frac{2xy}{r^2}$$

Multiplying by r^2,

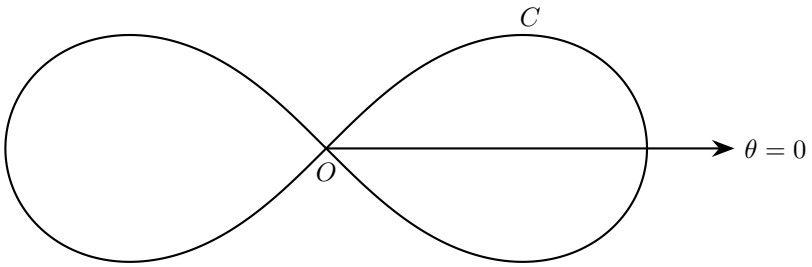
$$r^3 = r^2 + 2xy$$

Since $r^2 = x^2 + y^2$,

$$r^3 = x^2 + y^2 + 2xy = (x + y)^2$$

Also $r = \sqrt{x^2 + y^2}$, so the Cartesian equation is

$$(x^2 + y^2)^{3/2} = (x + y)^2$$



2. The diagram above shows the curve C with polar equation $r^2 = 8\cos 2\theta$.

(a) Show that a Cartesian equation of C is

$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$ **[4]**

(b) Show that the coordinate axes are lines of symmetry of C. **[2]**

(c) Find the exact area enclosed by each loop of C. **[4]**

Solution

(a) Using

$$x = r\cos\theta, \qquad y = r\sin\theta$$

we have

$$r^2 = x^2 + y^2$$

Also,

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

and since

$$\cos\theta = \frac{x}{r}, \qquad \sin\theta = \frac{y}{r}$$

it follows that

$$\cos 2\theta = \left(\frac{x}{r}\right)^2 - \left(\frac{y}{r}\right)^2$$
$$= \frac{x^2 - y^2}{r^2}$$

Now substitute into the polar equation $r^2 = 8\cos 2\theta$:

$$r^2 = 8 \cdot \frac{x^2 - y^2}{r^2}$$

and since $r^2 = x^2 + y^2$,

$$x^2 + y^2 = \frac{8(x^2 - y^2)}{x^2 + y^2}$$

Multiplying both sides by $x^2 + y^2$ gives

$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

(b) To test symmetry in the x-axis, replace y by $-y$:

$$(x^2 + (-y)^2)^2 = 8(x^2 - (-y)^2)$$
$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

which is unchanged, so the curve is symmetric about the x-axis.

To test symmetry in the y-axis, replace x by $-x$:

$$((-x)^2 + y^2)^2 = 8((-x)^2 - y^2)$$
$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

which is also unchanged, so the curve is symmetric about the y-axis.

Hence the coordinate axes are lines of symmetry of C.

(c) For a polar curve, the area enclosed is

$$A = \frac{1}{2}\int r^2 \, \mathrm{d}\theta$$

Here

$$r^2 = 8\cos 2\theta$$

One loop is traced when $r^2 \geq 0$ and θ runs from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$, since at the endpoints

$$\cos 2\theta = 0$$

so the curve passes through the pole.

Therefore the area of one loop is

$$\begin{aligned} A &= \frac{1}{2}\int_{-\pi/4}^{\pi/4} 8\cos 2\theta \, \mathrm{d}\theta \\ &= 4\int_{-\pi/4}^{\pi/4} \cos 2\theta \, \mathrm{d}\theta \\ &= 4\left[\frac{1}{2}\sin 2\theta\right]_{-\pi/4}^{\pi/4} \\ &= 2[\sin 2\theta]_{-\pi/4}^{\pi/4} \\ &= 2\left(\sin\frac{\pi}{2} - \sin\left(-\frac{\pi}{2}\right)\right) \\ &= 2(1 - (-1)) \\ &= 4 \end{aligned}$$

3. **(a)** On the same diagram, sketch the curve C_1 with polar equation

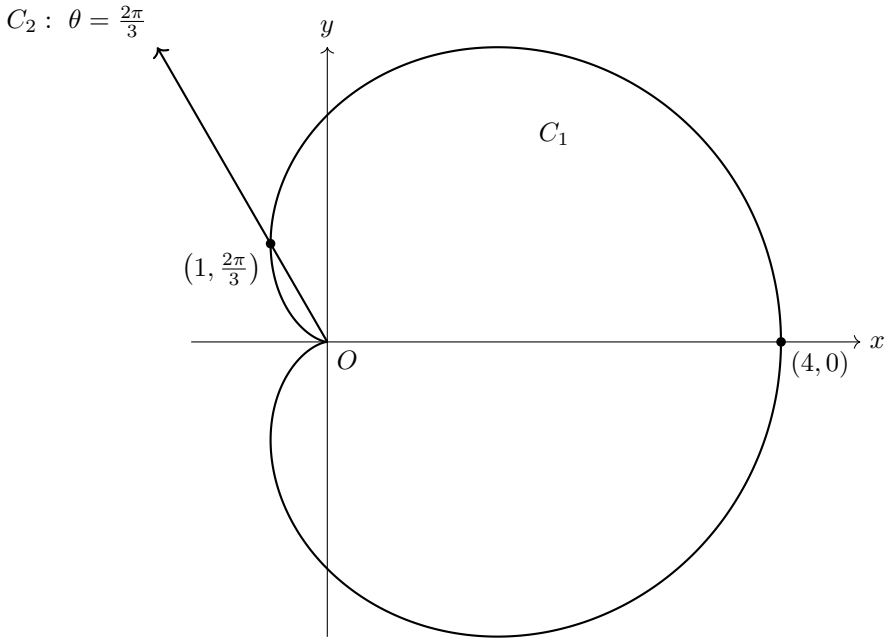
$$r = 2(1 + \cos\theta), \quad 0 \leq \theta \leq 2\pi$$

and the curve C_2 with polar equation $\theta = \frac{2\pi}{3}$. **[3]**

(b) Find the area of the smaller region bounded by C_1 and C_2. **[4]**

Solution

(a) The sketch should look like:



C_1 is a cardioid, symmetric about the initial line, with $r = 4$ at $\theta = 0$ and a cusp at the pole at $\theta = \pi$. C_2 is the half-line from the pole at angle $\frac{2\pi}{3}$ to the initial line. Substituting $\theta = \frac{2\pi}{3}$ into C_1:

$$r = 2\left(1 + \cos\frac{2\pi}{3}\right) = 2\left(1 - \frac{1}{2}\right) = 1$$

So the curves meet at the pole and at $\left(1, \frac{2\pi}{3}\right)$.

(b) The smaller region is the part inside the cardioid between

$$\theta = \frac{2\pi}{3} \quad \text{and} \quad \theta = \pi$$

Using the polar area formula

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, \mathrm{d}\theta$$

we get

$$A = \frac{1}{2}\int_{2\pi/3}^{\pi} [2(1 + \cos\theta)]^2 \, \mathrm{d}\theta$$

Now simplify the integrand:

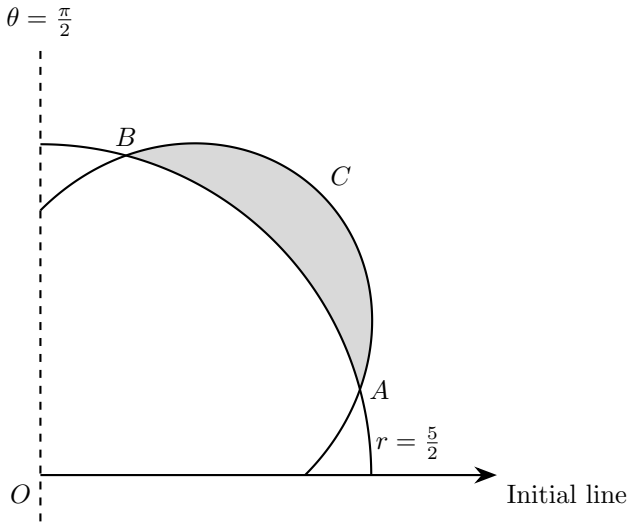
$$\begin{aligned} A &= 2\int_{2\pi/3}^{\pi} (1 + 2\cos\theta + \cos^2\theta)\,\mathrm{d}\theta \\ &= 2\int_{2\pi/3}^{\pi} \left(1 + 2\cos\theta + \frac{1+\cos 2\theta}{2}\right)\mathrm{d}\theta \\ &= \int_{2\pi/3}^{\pi} (3 + 4\cos\theta + \cos 2\theta)\,\mathrm{d}\theta \end{aligned}$$

Integrating,

$$\begin{aligned} A &= \left[3\theta + 4\sin\theta + \frac{1}{2}\sin 2\theta\right]_{2\pi/3}^{\pi} \\ &= \left(3\pi + 4\sin\pi + \frac{1}{2}\sin 2\pi\right) - \left(3\cdot\frac{2\pi}{3} + 4\sin\frac{2\pi}{3} + \frac{1}{2}\sin\frac{4\pi}{3}\right) \\ &= 3\pi - \left(2\pi + 4\cdot\frac{\sqrt{3}}{2} + \frac{1}{2}\cdot\left(-\frac{\sqrt{3}}{2}\right)\right) \\ &= \pi - 2\sqrt{3} + \frac{\sqrt{3}}{4} \\ &= \pi - \frac{7\sqrt{3}}{4} \end{aligned}$$

So the area of the smaller region is

$$\pi - \frac{7\sqrt{3}}{4} \approx 0.111$$



4. The diagram shows part of the curve C with polar equation

$$r = 2 + \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

and the circle

$$r = \frac{5}{2}$$

The curve and the circle intersect at the points A and B, as shown.

Find, correct to three significant figures, the area of the shaded region enclosed by C and the circle. **[6]**

Solution

For the intersection points of the curve and the circle, set the two values of r equal:

$$2 + \sin 2\theta = \frac{5}{2}$$

So

$$\sin 2\theta = \frac{1}{2}$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we have $0 \leq 2\theta \leq \pi$, so the solutions are

$$2\theta = \frac{\pi}{6}, \ \frac{5\pi}{6}$$

Hence

$$\theta = \frac{\pi}{12}, \ \frac{5\pi}{12}$$

So the points of intersection are at $\theta = \frac{\pi}{12}$ and $\theta = \frac{5\pi}{12}$.
For $\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$, we have $\sin 2\theta \geq \frac{1}{2}$, so

$$2 + \sin 2\theta \geq \frac{5}{2}$$

Therefore the curve C lies outside the circle on this interval, and the required area is

$$A = \frac{1}{2}\int_{\pi/12}^{5\pi/12}\left[(2+\sin 2\theta)^2 - \left(\frac{5}{2}\right)^2\right]\mathrm{d}\theta$$

Now expand the integrand:

$$\begin{aligned} A &= \frac{1}{2}\int_{\pi/12}^{5\pi/12}\left(4 + 4\sin 2\theta + \sin^2 2\theta - \frac{25}{4}\right)\mathrm{d}\theta \\ &= \frac{1}{2}\int_{\pi/12}^{5\pi/12}\left(-\frac{9}{4} + 4\sin 2\theta + \sin^2 2\theta\right)\mathrm{d}\theta \end{aligned}$$

Use the identity

$$\sin^2 2\theta = \frac{1-\cos 4\theta}{2}$$

Then

$$\begin{aligned} A &= \frac{1}{2}\int_{\pi/12}^{5\pi/12}\left(-\frac{9}{4} + 4\sin 2\theta + \frac{1-\cos 4\theta}{2}\right)\mathrm{d}\theta \\ &= \frac{1}{2}\int_{\pi/12}^{5\pi/12}\left(-\frac{7}{4} + 4\sin 2\theta - \frac{1}{2}\cos 4\theta\right)\mathrm{d}\theta \end{aligned}$$

Integrating,

$$A = \left[-\frac{7\theta}{8} - \cos 2\theta - \frac{1}{16}\sin 4\theta\right]_{\pi/12}^{5\pi/12}$$

Now substitute the limits:

$$A = \left(-\frac{35\pi}{96} - \cos\frac{5\pi}{6} - \frac{1}{16}\sin\frac{5\pi}{3}\right) - \left(-\frac{7\pi}{96} - \cos\frac{\pi}{6} - \frac{1}{16}\sin\frac{\pi}{3}\right)$$

Using

$$\cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \quad \sin\frac{5\pi}{3} = -\frac{\sqrt{3}}{2}, \quad \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}, \quad \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

gives

$$\begin{aligned} A &= \left(-\frac{35\pi}{96} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{32}\right) - \left(-\frac{7\pi}{96} - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{32}\right) \\ &= -\frac{28\pi}{96} + \frac{34\sqrt{3}}{32} \\ &= -\frac{7\pi}{24} + \frac{17\sqrt{3}}{16} \end{aligned}$$

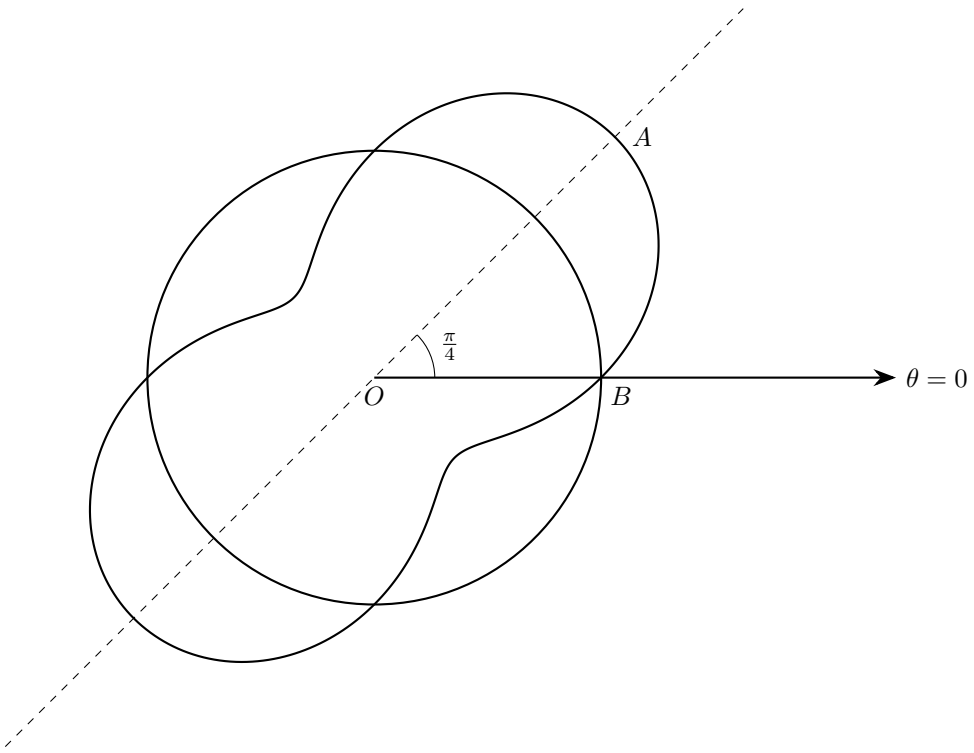
So the exact area is

$$A = \frac{17\sqrt{3}}{16} - \frac{7\pi}{24}$$

and numerically

$$A \approx 0.924$$

Therefore, the area of the shaded region is 0.924 square units, correct to three significant figures.



5. The diagram above shows the curve C with polar equation $r = a(2 + \sin 2\theta)$ and the circle $r = 2a$ for $-\pi \leq \theta \leq \pi$, where a is a positive constant.

(a) Write down the polar coordinates of the points A and B. **[2]**

(b) Explain why C is symmetrical about the line $\theta = \frac{\pi}{4}$. **[2]**

(c) Find, in terms of a, the exact total area of the regions that lie inside C and outside the circle. **[4]**

Solution

(a) For point A, the diagram shows $\theta = \frac{\pi}{4}$. Then

$$r = a\left(2 + \sin\frac{\pi}{2}\right) = a(2 + 1) = 3a$$

so

$$A = \left(3a, \frac{\pi}{4}\right)$$

For point B, $\theta = 0$, so

$$r = a(2 + \sin 0) = 2a$$

hence

$$B = (2a, 0)$$

(b) A curve is symmetrical about the line $\theta = \frac{\pi}{4}$ if replacing θ by

$$\frac{\pi}{2} - \theta$$

leaves the equation unchanged, since reflection in the line $\theta = \frac{\pi}{4}$ sends θ to $\frac{\pi}{2} - \theta$.

Here,

$$r = a\left(2 + \sin 2\left(\frac{\pi}{2} - \theta\right)\right) = a(2 + \sin(\pi - 2\theta))$$

and since $\sin(\pi - 2\theta) = \sin 2\theta$,

$$r = a(2 + \sin 2\theta)$$

So the equation is unchanged, and therefore C is symmetrical about the line $\theta = \frac{\pi}{4}$.

(c) The region is inside C and outside the circle $r = 2a$, so we need

$$a(2 + \sin 2\theta) \geq 2a$$

Since $a > 0$, this gives

$$2 + \sin 2\theta \geq 2$$

$$\sin 2\theta \geq 0$$

On $-\pi \leq \theta \leq \pi$, this happens on

$$-\pi \leq \theta \leq -\frac{\pi}{2} \qquad \text{and} \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

so there are two equal regions.

Find the area of one region, for $0 \leq \theta \leq \frac{\pi}{2}$.

Using the polar area formula,

$$\text{Area} = \frac{1}{2}\int r^2 \, \mathrm{d}\theta$$

the required area for one region is

$$A_1 = \frac{1}{2}\int_0^{\pi/2} \left(a^2(2 + \sin 2\theta)^2 - (2a)^2\right) \mathrm{d}\theta$$

$$A_1 = \frac{a^2}{2}\int_0^{\pi/2} \left((2 + \sin 2\theta)^2 - 4\right) \mathrm{d}\theta$$

Expand the bracket:

$$(2 + \sin 2\theta)^2 - 4 = 4 + 4\sin 2\theta + \sin^2 2\theta - 4 = 4\sin 2\theta + \sin^2 2\theta$$

So

$$A_1 = \frac{a^2}{2}\int_0^{\pi/2} \left(4\sin 2\theta + \sin^2 2\theta\right) \mathrm{d}\theta$$

Now evaluate each part:

$$\int_0^{\pi/2} 4\sin 2\theta \, \mathrm{d}\theta = \left[-2\cos 2\theta\right]_0^{\pi/2} = 2 - (-2) = 4$$

Also,

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

so

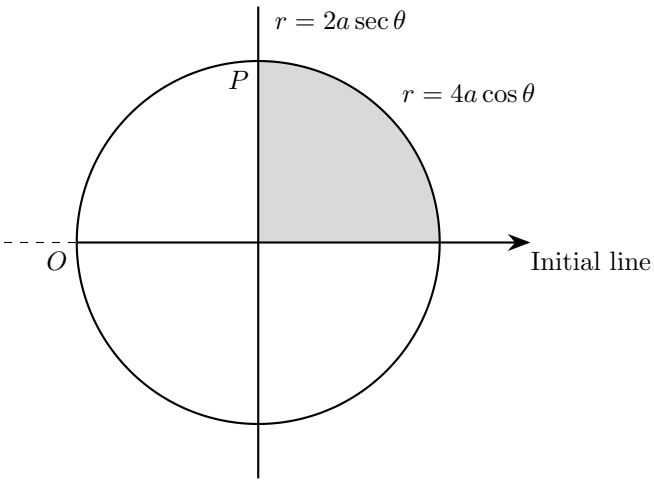
$$\int_0^{\pi/2} \sin^2 2\theta \, \mathrm{d}\theta = \int_0^{\pi/2} \frac{1-\cos 4\theta}{2} \, \mathrm{d}\theta = \left[\frac{\theta}{2} - \frac{\sin 4\theta}{8}\right]_0^{\pi/2} = \frac{\pi}{4}$$

Therefore

$$A_1 = \frac{a^2}{2}\left(4 + \frac{\pi}{4}\right) = a^2\left(2 + \frac{\pi}{8}\right)$$

There are two such regions, so the total area is

$$2A_1 = 2a^2\left(2 + \frac{\pi}{8}\right) = a^2\left(4 + \frac{\pi}{4}\right) = \frac{a^2(16+\pi)}{4}$$



6. The diagram shows a sketch of the curves with polar equations

$$r = 4a\cos\theta \text{ and } r = 2a\sec\theta, \quad a > 0$$

(a) Find the polar coordinates of the point of intersection P of the two curves in the first quadrant. **[4]**

(b) Find the area, shaded in the figure, bounded by the two curves and by the initial line $\theta = 0$, giving your answer in terms of a and π. **[4]**

Solution

(a) At a point of intersection, the two expressions for r are equal:

$$4a\cos\theta = 2a\sec\theta$$

Since $a > 0$, divide by $2a$:

$$2\cos\theta = \sec\theta = \frac{1}{\cos\theta}$$

So

$$2\cos^2\theta = 1$$

$$\cos^2\theta = \frac{1}{2}$$

In the first quadrant, $\cos\theta > 0$, so

$$\cos\theta = \frac{1}{\sqrt{2}}$$

hence

$$\theta = \frac{\pi}{4}$$

Now find r:

$$r = 4a\cos\frac{\pi}{4} = 4a\cdot\frac{1}{\sqrt{2}} = 2\sqrt{2}\,a$$

So the polar coordinates of P are

$$\left(2\sqrt{2}\,a, \frac{\pi}{4}\right)$$

(b) The required shaded area is between the curves from $\theta = 0$ to the intersection at $\theta = \frac{\pi}{4}$.

For $0 \leq \theta \leq \frac{\pi}{4}$, the outer curve is

$$r = 4a\cos\theta$$

and the inner curve is

$$r = 2a\sec\theta$$

So, using

$$\text{Area} = \frac{1}{2}\int_{\alpha}^{\beta}\left(r_{\text{outer}}^2 - r_{\text{inner}}^2\right)\mathrm{d}\theta$$

we get

$$A = \frac{1}{2}\int_{0}^{\pi/4}\left((4a\cos\theta)^2 - (2a\sec\theta)^2\right)\mathrm{d}\theta$$

Simplify:

$$A = \frac{1}{2}\int_{0}^{\pi/4}\left(16a^2\cos^2\theta - 4a^2\sec^2\theta\right)\mathrm{d}\theta$$

$$A = 8a^2\int_{0}^{\pi/4}\cos^2\theta\,\mathrm{d}\theta - 2a^2\int_{0}^{\pi/4}\sec^2\theta\,\mathrm{d}\theta$$

Use

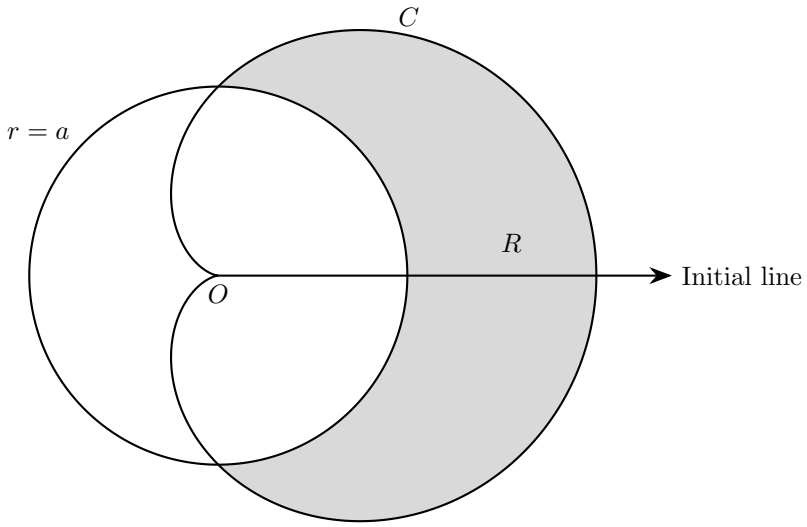
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

so

$$\int\cos^2\theta\,\mathrm{d}\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4}$$

Therefore

$$\begin{aligned} A &= 8a^2\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\pi/4} - 2a^2[\tan\theta]_0^{\pi/4} \\ &= 8a^2\left(\frac{\pi}{8} + \frac{1}{4}\right) - 2a^2(1) \\ &= \pi a^2 + 2a^2 - 2a^2 \\ &= \pi a^2 \end{aligned}$$



7. The figure above shows a sketch of the curve C with polar equation

$$r = a(1 + \cos\theta)$$

and the circle

$$r = a$$

where $a > 0$ and $-\pi \leq \theta \leq \pi$.

The shaded region inside C and outside the circle has area

$$18 + \frac{9\pi}{4}$$

Find the value of a. **[8]**

Solution

For the shaded region, we first find where the curve and the circle meet.

The curve is

$$r = a(1 + \cos\theta)$$

and the circle is

$$r = a$$

At points of intersection,

$$a(1 + \cos\theta) = a$$

Since $a > 0$, divide by a:

$$1 + \cos\theta = 1$$

so

$$\cos\theta = 0$$

Hence

$$\theta = \pm\frac{\pi}{2}$$

So the shaded region is between $\theta = -\frac{\pi}{2}$ and $\theta = \frac{\pi}{2}$, with outer radius

$$r = a(1 + \cos\theta)$$

and inner radius

$$r = a$$

Using the polar area formula,

$$\text{Area} = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, \mathrm{d}\theta$$

the shaded area is

$$\frac{1}{2}\int_{-\pi/2}^{\pi/2} \left([a(1 + \cos\theta)]^2 - a^2\right) \mathrm{d}\theta$$

Expand the bracket:

$$\begin{aligned}[a(1 + \cos\theta)]^2 - a^2 &= a^2(1 + 2\cos\theta + \cos^2\theta) - a^2 \\ &= a^2(2\cos\theta + \cos^2\theta)\end{aligned}$$

So

$$\text{Area} = \frac{a^2}{2}\int_{-\pi/2}^{\pi/2} \left(2\cos\theta + \cos^2\theta\right) \mathrm{d}\theta$$

Now evaluate the two parts separately.

First,

$$\int_{-\pi/2}^{\pi/2} 2\cos\theta \, \mathrm{d}\theta = 2[\sin\theta]_{-\pi/2}^{\pi/2} = 2(1 - (-1)) = 4$$

For the second part, use

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

So

$$\begin{aligned}\int_{-\pi/2}^{\pi/2} \cos^2\theta \, \mathrm{d}\theta &= \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2\theta}{2} \, \mathrm{d}\theta \\ &= \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_{-\pi/2}^{\pi/2} \\ &= \left(\frac{\pi}{4} + 0\right) - \left(-\frac{\pi}{4} + 0\right) \\ &= \frac{\pi}{2}\end{aligned}$$

Therefore

$$\text{Area} = \frac{a^2}{2}\left(4 + \frac{\pi}{2}\right)$$

$$= 2a^2 + \frac{\pi a^2}{4}$$

We are told this area is

$$18 + \frac{9\pi}{4}$$

So

$$2a^2 + \frac{\pi a^2}{4} = 18 + \frac{9\pi}{4}$$

Multiply through by 4:

$$8a^2 + \pi a^2 = 72 + 9\pi$$

$$a^2(8 + \pi) = 9(8 + \pi)$$

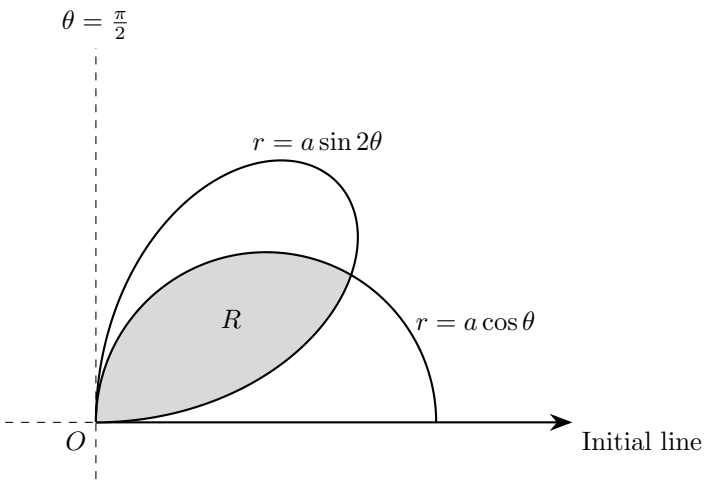
Hence

$$a^2 = 9$$

Since $a > 0$,

$$a = 3$$

Therefore, the value of a is 3.



8. In the first quadrant, the curve $r = a\sin 2\theta$ intersects the circle $r = a\cos\theta$ at the pole and at one other point.

The region R shown consists of the points that lie inside both curves.

Find the exact area of R in terms of a. **[7]**

Solution

For the points of intersection, set the two expressions for r equal:

$$a\sin 2\theta = a\cos\theta$$

Since $a \neq 0$,

$$\sin 2\theta = \cos\theta$$

Using $\sin 2\theta = 2\sin\theta\cos\theta$,

$$2\sin\theta\cos\theta = \cos\theta$$

$$\cos\theta(2\sin\theta - 1) = 0$$

In the first quadrant, $0 \leq \theta \leq \frac{\pi}{2}$, so

$$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

or

$$2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

At $\theta = \frac{\pi}{2}$,

$$r = a\cos\frac{\pi}{2} = 0$$

so this is the pole.

At the other intersection,

$$r = a\cos\frac{\pi}{6} = \frac{a\sqrt{3}}{2}$$

So the non-pole intersection is at

$$(r, \theta) = \left(\frac{a\sqrt{3}}{2}, \frac{\pi}{6} \right)$$

Now find which curve forms the boundary of the region inside both curves.

For $0 \leq \theta \leq \frac{\pi}{6}$,

$$a \sin 2\theta \leq a \cos \theta$$

so the smaller radius is $r = a \sin 2\theta$.

For $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$,

$$a \cos \theta \leq a \sin 2\theta$$

so the smaller radius is $r = a \cos \theta$.

Hence, using the polar area formula

$$\text{Area} = \frac{1}{2} \int r^2 \, \mathrm{d}\theta$$

the required area is

$$\text{Area}(R) = \frac{1}{2} a^2 \left(\int_0^{\pi/6} \sin^2 2\theta \, \mathrm{d}\theta + \int_{\pi/6}^{\pi/2} \cos^2 \theta \, \mathrm{d}\theta \right)$$

Now evaluate the integrals.

First, use

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

Then

$$\begin{aligned} \int_0^{\pi/6} \sin^2 2\theta \, \mathrm{d}\theta &= \int_0^{\pi/6} \frac{1 - \cos 4\theta}{2} \, \mathrm{d}\theta \\ &= \left[\frac{\theta}{2} - \frac{\sin 4\theta}{8} \right]_0^{\pi/6} \\ &= \frac{\pi}{12} - \frac{\sin(2\pi/3)}{8} \\ &= \frac{\pi}{12} - \frac{\sqrt{3}}{16} \end{aligned}$$

Next, use

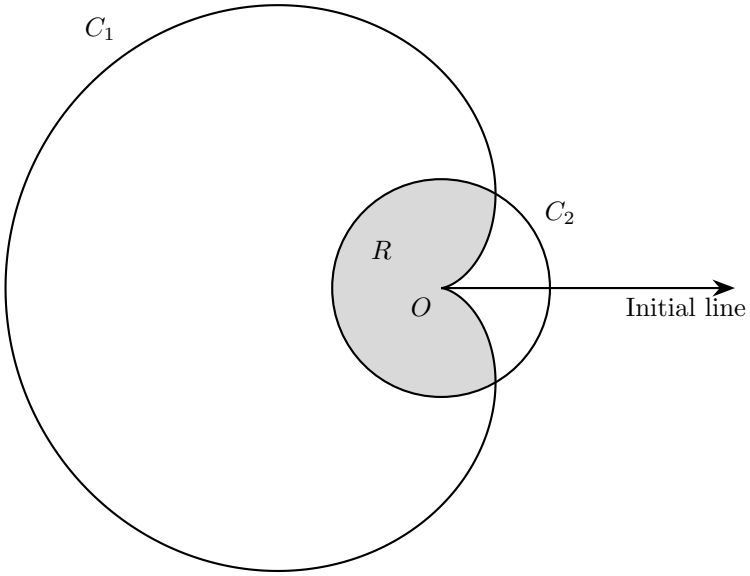
$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

Then

$$\begin{aligned}\int_{\pi/6}^{\pi/2} \cos^2\theta \,\mathrm{d}\theta &= \int_{\pi/6}^{\pi/2} \frac{1+\cos 2\theta}{2}\,\mathrm{d}\theta \\ &= \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_{\pi/6}^{\pi/2} \\ &= \left(\frac{\pi}{4} + 0\right) - \left(\frac{\pi}{12} + \frac{\sin(\pi/3)}{4}\right) \\ &= \frac{\pi}{6} - \frac{\sqrt{3}}{8}\end{aligned}$$

Substitute back:

$$\begin{aligned}\text{Area}(R) &= \frac{1}{2}a^2\left(\frac{\pi}{12} - \frac{\sqrt{3}}{16} + \frac{\pi}{6} - \frac{\sqrt{3}}{8}\right) \\ &= \frac{1}{2}a^2\left(\frac{\pi}{4} - \frac{3\sqrt{3}}{16}\right) \\ &= \frac{a^2(4\pi - 3\sqrt{3})}{32}\end{aligned}$$



9. The diagram above shows the curve C_1 with polar equation $r = 2a(1 - \cos\theta)$, $0 \leq \theta \leq 2\pi$, and the circle C_2 with polar equation $r = a$, $0 \leq \theta \leq 2\pi$, where a is a positive constant.

(a) Find, in terms of a, the polar coordinates of the points where C_1 meets C_2. **[3]**

The common region enclosed by C_1 and C_2 is shaded in the diagram above.

(b) Find the area of the shaded region R, giving your answer in the form

$$\frac{1}{12}a^2\left(p\pi + q\sqrt{3}\right)$$

where p and q are integers to be found. **[5]**

Solution

(a) At the points of intersection, the two values of r are equal, so

$$2a(1 - \cos\theta) = a$$

Since $a > 0$, divide through by a:

$$2(1 - \cos\theta) = 1$$

$$2 - 2\cos\theta = 1$$

$$\cos\theta = \frac{1}{2}$$

For $0 \leq \theta \leq 2\pi$, this gives

$$\theta = \frac{\pi}{3},\ \frac{5\pi}{3}$$

Substituting into either curve gives

$$r = a$$

So the points where C_1 meets C_2 are

$$(a, \tfrac{\pi}{3}) \quad \text{and} \quad (a, \tfrac{5\pi}{3})$$

(b) For the common region, the boundary is given by the smaller of the two radii.

Compare the curves:

$$2a(1-\cos\theta) \leq a$$

$$2(1-\cos\theta) \leq 1$$

$$\cos\theta \geq \frac{1}{2}$$

So:

$$0 \leq \theta \leq \frac{\pi}{3} \quad \text{and} \quad \frac{5\pi}{3} \leq \theta \leq 2\pi$$

give the cardioid $r = 2a(1-\cos\theta)$ as the inner curve, and

$$\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

gives the circle $r = a$ as the inner curve.

Using the polar area formula

$$A = \frac{1}{2}\int r^2 \, \mathrm{d}\theta$$

the shaded area is

$$A = \frac{1}{2}\left[\int_0^{\pi/3} \big(2a(1-\cos\theta)\big)^2 \, \mathrm{d}\theta + \int_{\pi/3}^{5\pi/3} a^2 \, \mathrm{d}\theta + \int_{5\pi/3}^{2\pi} \big(2a(1-\cos\theta)\big)^2 \, \mathrm{d}\theta\right]$$

By symmetry, the first and third integrals are equal, so

$$\begin{aligned} A &= \int_0^{\pi/3} \big(2a(1-\cos\theta)\big)^2 \, \mathrm{d}\theta + \frac{1}{2}\int_{\pi/3}^{5\pi/3} a^2 \, \mathrm{d}\theta \\ &= 4a^2 \int_0^{\pi/3} (1 - 2\cos\theta + \cos^2\theta) \, \mathrm{d}\theta + \frac{1}{2}a^2\left(\frac{5\pi}{3} - \frac{\pi}{3}\right) \end{aligned}$$

Now use

$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

So

$$\begin{aligned} A &= 4a^2 \int_0^{\pi/3} \left(1 - 2\cos\theta + \frac{1+\cos 2\theta}{2}\right) \mathrm{d}\theta + \frac{1}{2}a^2 \cdot \frac{4\pi}{3} \\ &= 4a^2 \int_0^{\pi/3} \left(\frac{3}{2} - 2\cos\theta + \frac{1}{2}\cos 2\theta\right) \mathrm{d}\theta + \frac{2\pi a^2}{3} \end{aligned}$$

Integrating,

$$\begin{aligned}
A &= 4a^2\left[\frac{3}{2}\theta - 2\sin\theta + \frac{1}{4}\sin 2\theta\right]_0^{\pi/3} + \frac{2\pi a^2}{3} \\
&= 4a^2\left(\frac{3}{2}\cdot\frac{\pi}{3} - 2\sin\frac{\pi}{3} + \frac{1}{4}\sin\frac{2\pi}{3}\right) + \frac{2\pi a^2}{3} \\
&= 4a^2\left(\frac{\pi}{2} - 2\cdot\frac{\sqrt{3}}{2} + \frac{1}{4}\cdot\frac{\sqrt{3}}{2}\right) + \frac{2\pi a^2}{3} \\
&= 4a^2\left(\frac{\pi}{2} - \sqrt{3} + \frac{\sqrt{3}}{8}\right) + \frac{2\pi a^2}{3} \\
&= 4a^2\left(\frac{\pi}{2} - \frac{7\sqrt{3}}{8}\right) + \frac{2\pi a^2}{3} \\
&= 2\pi a^2 - \frac{7\sqrt{3}}{2}a^2 + \frac{2\pi a^2}{3} \\
&= \frac{8\pi a^2}{3} - \frac{7\sqrt{3}\,a^2}{2}
\end{aligned}$$

Putting this over a denominator of 12,

$$A = \frac{1}{12}a^2(32\pi - 42\sqrt{3})$$

Hence the shaded area is

$$\frac{1}{12}a^2(32\pi - 42\sqrt{3})$$

so

$$p = 32, \qquad q = -42$$

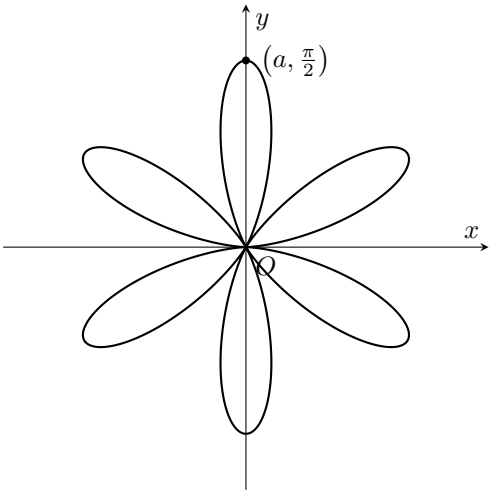
10. A curve has polar equation $r = a\sin^2 3\theta$, where $a > 0$ and $0 \leq \theta < 2\pi$.

(a) Sketch the curve. **[2]**

(b) Find, in exact form, the total area of the regions enclosed by the curve. **[6]**

Solution

(a) The sketch should look like:



Since $r = a\sin^2 3\theta \geq 0$ for all θ, every θ gives a point on the curve. It passes through the pole when $\sin 3\theta = 0$, that is at $\theta = \frac{n\pi}{3}$ for $n = 0, 1, \ldots, 5$, so the curve is a rose with 6 petals, one between each pair of consecutive zeros. The maximum value of r is a, when $\sin^2 3\theta = 1$, that is at the petal tips $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$.

(b) The polar area formula is

$$A = \frac{1}{2}\int r^2 \,\mathrm{d}\theta$$

One petal is traced out as θ goes from 0 to $\frac{\pi}{3}$, since $r = 0$ at both endpoints and $r > 0$ between them.

So the area of one petal is

$$A_1 = \frac{1}{2}\int_0^{\pi/3} a^2 \sin^4(3\theta)\,\mathrm{d}\theta$$

Use

$$\sin^4 x = \left(\frac{1-\cos 2x}{2}\right)^2 = \frac{1 - 2\cos 2x + \cos^2 2x}{4} = \frac{1 - 2\cos 2x + \frac{1+\cos 4x}{2}}{4} = \frac{3 - 4\cos 2x + \cos 4x}{8}$$

Therefore

$$\sin^4(3\theta) = \frac{3 - 4\cos 6\theta + \cos 12\theta}{8}$$

Substitute into the integral:

$$\begin{aligned} A_1 &= \frac{a^2}{2}\int_0^{\pi/3} \frac{3 - 4\cos 6\theta + \cos 12\theta}{8}\,\mathrm{d}\theta \\ &= \frac{a^2}{16}\int_0^{\pi/3} (3 - 4\cos 6\theta + \cos 12\theta)\,\mathrm{d}\theta \\ &= \frac{a^2}{16}\left[3\theta - \frac{4}{6}\sin 6\theta + \frac{1}{12}\sin 12\theta\right]_0^{\pi/3} \end{aligned}$$

Now evaluate:

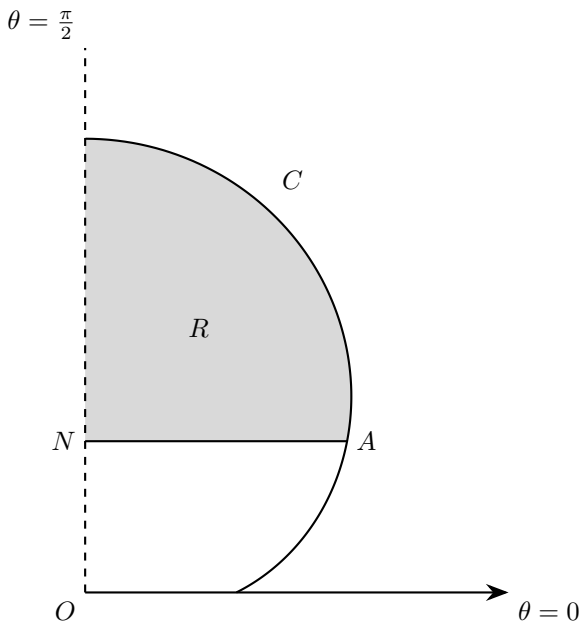
$$\sin\left(6\cdot\frac{\pi}{3}\right) = \sin 2\pi = 0, \qquad \sin\left(12\cdot\frac{\pi}{3}\right) = \sin 4\pi = 0$$

Hence

$$\begin{aligned} A_1 &= \frac{a^2}{16}\left[3\cdot\frac{\pi}{3} - 0 + 0\right] \\ &= \frac{\pi a^2}{16} \end{aligned}$$

There are 6 identical petals, so the total area is

$$\begin{aligned} A_{\text{total}} &= 6\cdot\frac{\pi a^2}{16} \\ &= \frac{3\pi a^2}{8} \end{aligned}$$



11. The curve C shown in the diagram above has polar equation

$$r = 1 + 2\sin\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The point A on C has y-coordinate 1.

The point N lies on the line $\theta = \frac{\pi}{2}$ and AN is perpendicular to the line $\theta = \frac{\pi}{2}$.

The finite region R, shown shaded in the diagram above, is bounded by the curve C, the line $\theta = \frac{\pi}{2}$ and the line AN.

Find the exact area of the shaded region R. **[7]**

Solution

For the curve

$$r = 1 + 2\sin\theta, \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

the point A has y-coordinate 1.

In polar coordinates,

$$y = r\sin\theta$$

so at A,

$$(1 + 2\sin\theta)\sin\theta = 1$$

Hence

$$2\sin^2\theta + \sin\theta - 1 = 0$$

$$(2\sin\theta - 1)(\sin\theta + 1) = 0$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, we take

$$\sin\theta = \frac{1}{2}$$

so

$$\theta_A = \frac{\pi}{6}$$

Then

$$r_A = 1 + 2\sin\frac{\pi}{6} = 1 + 2\left(\frac{1}{2}\right) = 2$$

Therefore

$$A = \left(2, \frac{\pi}{6}\right)$$

and in Cartesian coordinates

$$x = 2\cos\frac{\pi}{6} = \sqrt{3}, \qquad y = 2\sin\frac{\pi}{6} = 1$$

so

$$A = (\sqrt{3}, 1)$$

The line $\theta = \frac{\pi}{2}$ is the y-axis, so the line AN, perpendicular to it, is horizontal. Since it passes through A, its equation is

$$y = 1$$

Now consider the shaded region R. For angles

$$\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$$

the outer boundary is the curve

$$r = 1 + 2\sin\theta$$

and the inner boundary is the line $y = 1$.

Using $y = r\sin\theta$, the line $y = 1$ becomes

$$r\sin\theta = 1$$

so

$$r = \frac{1}{\sin\theta} = \operatorname{cosec}\theta$$

Using the polar area formula,

$$\text{Area}(R) = \frac{1}{2}\int_{\pi/6}^{\pi/2}\left[\left(1 + 2\sin\theta\right)^2 - \operatorname{cosec}^2\theta\right]\mathrm{d}\theta$$

Expand the bracket:

$$\text{Area}(R) = \frac{1}{2}\int_{\pi/6}^{\pi/2}\left(1 + 4\sin\theta + 4\sin^2\theta - \operatorname{cosec}^2\theta\right)\mathrm{d}\theta$$

Now integrate term by term. Using

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

we get

$$\begin{aligned}\int \big(1+4\sin\theta+4\sin^2\theta-\operatorname{cosec}^2\theta\big)\mathrm{d}\theta &= \int \big(1+4\sin\theta+2-2\cos 2\theta-\operatorname{cosec}^2\theta\big)\mathrm{d}\theta \\ &= 3\theta-4\cos\theta-\sin 2\theta+\cot\theta\end{aligned}$$

So

$$\text{Area}(R) = \frac{1}{2}[3\theta-4\cos\theta-\sin 2\theta+\cot\theta]_{\pi/6}^{\pi/2}$$

Now substitute the limits.

At $\theta = \frac{\pi}{2}$:

$$3\left(\frac{\pi}{2}\right)-4\cos\frac{\pi}{2}-\sin\pi+\cot\frac{\pi}{2} = \frac{3\pi}{2}$$

At $\theta = \frac{\pi}{6}$:

$$\begin{aligned}3\left(\frac{\pi}{6}\right)-4\cos\frac{\pi}{6}-\sin\frac{\pi}{3}+\cot\frac{\pi}{6} &= \frac{\pi}{2}-4\left(\frac{\sqrt{3}}{2}\right)-\frac{\sqrt{3}}{2}+\sqrt{3} \\ &= \frac{\pi}{2}-2\sqrt{3}-\frac{\sqrt{3}}{2}+\sqrt{3} \\ &= \frac{\pi}{2}-\frac{3\sqrt{3}}{2}\end{aligned}$$

Therefore

$$\begin{aligned}\text{Area}(R) &= \frac{1}{2}\left(\frac{3\pi}{2}-\left(\frac{\pi}{2}-\frac{3\sqrt{3}}{2}\right)\right) \\ &= \frac{1}{2}\left(\pi+\frac{3\sqrt{3}}{2}\right) \\ &= \frac{\pi}{2}+\frac{3\sqrt{3}}{4}\end{aligned}$$

12. The curve C has polar equation

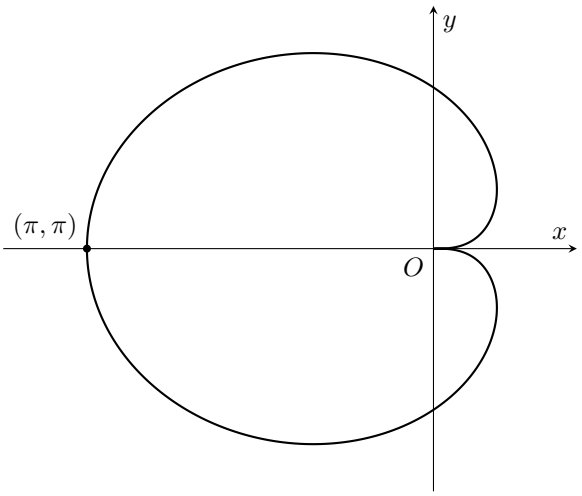
$$r^2 = \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \qquad 0 \leq \theta \leq 2\pi$$

(a) Sketch C and state the polar coordinates of the point on C furthest from the pole. **[4]**

(b) Using the substitution $u = \theta - \pi$, or otherwise, find the exact area enclosed by C. **[6]**

Solution

(a) The sketch should look like:



When $\theta = 0$ or $\theta = 2\pi$, $r^2 = \frac{\pi^2 - \pi^2}{1+\pi^2} = 0$, so C starts and ends at the pole. Since r depends only on $(\theta - \pi)^2$, $r(2\pi - \theta) = r(\theta)$, so C is symmetric about the initial line.

For the furthest point, let

$$w = (\theta - \pi)^2$$

Then

$$r^2 = \frac{\pi^2 - w}{1 + w}, \qquad 0 \leq w \leq \pi^2$$

Differentiate with respect to w:

$$\frac{\mathrm{d}(r^2)}{\mathrm{d}w} = \frac{-(1+w) - (\pi^2 - w)}{(1+w)^2} = -\frac{\pi^2 + 1}{(1+w)^2} < 0$$

So r^2 decreases as w increases, and its maximum occurs when $w = 0$.

Hence

$$(\theta - \pi)^2 = 0 \implies \theta = \pi$$

and then

$$r^2 = \frac{\pi^2}{1} = \pi^2$$

Since we take $r > 0$, this gives

$$r = \pi$$

So the point on C furthest from the pole is

$$(\pi, \pi)$$

(b) The area enclosed by a polar curve is

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, \mathrm{d}\theta$$

So here

$$A = \frac{1}{2}\int_{0}^{2\pi} \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \, \mathrm{d}\theta$$

Use the substitution

$$u = \theta - \pi$$

Then $\mathrm{d}u = \mathrm{d}\theta$, and the limits change as follows:

$$\theta = 0 \implies u = -\pi, \qquad \theta = 2\pi \implies u = \pi$$

Hence

$$A = \frac{1}{2}\int_{-\pi}^{\pi} \frac{\pi^2 - u^2}{1 + u^2} \, \mathrm{d}u$$

Now simplify the integrand:

$$\begin{aligned} \frac{\pi^2 - u^2}{1 + u^2} &= \frac{-(1 + u^2) + (\pi^2 + 1)}{1 + u^2} \\ &= -1 + \frac{\pi^2 + 1}{1 + u^2} \end{aligned}$$

Therefore

$$\begin{aligned} A &= \frac{1}{2}\int_{-\pi}^{\pi} \left(-1 + \frac{\pi^2 + 1}{1 + u^2}\right) \mathrm{d}u \\ &= \frac{1}{2}\Big[-u + (\pi^2 + 1)\arctan u\Big]_{-\pi}^{\pi} \end{aligned}$$

using

$$\int \frac{1}{1 + u^2} \, \mathrm{d}u = \arctan u + c$$

Now substitute the limits:

$$\begin{aligned} A &= \frac{1}{2}\Big[\big(-\pi + (\pi^2 + 1)\arctan \pi\big) - \big(\pi - (\pi^2 + 1)\arctan \pi\big)\Big] \\ &= \frac{1}{2}\Big[-2\pi + 2(\pi^2 + 1)\arctan \pi\Big] \\ &= (\pi^2 + 1)\arctan(\pi) - \pi \end{aligned}$$